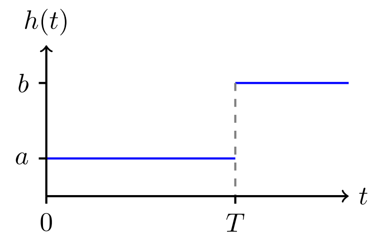
Q. 1 (Heart attack)**.** Here is a very unrealistic model of how a heart attack affects your life expectancy. The lifetime L of a person has the following hazard rate $h(t)$:



T is the time at which the person gets a heart attack. Before this time, the hazard rate $h(t) = a$ for $t < T$; after the heart attack, this increases to $h(t) = b$ for $t \geq T$. The time T at which a person gets a heart attack is itself random: $T \sim \text{Expon}(\mu)$.

a. What is the expected lifetime $\mathbf{E}(L)$?

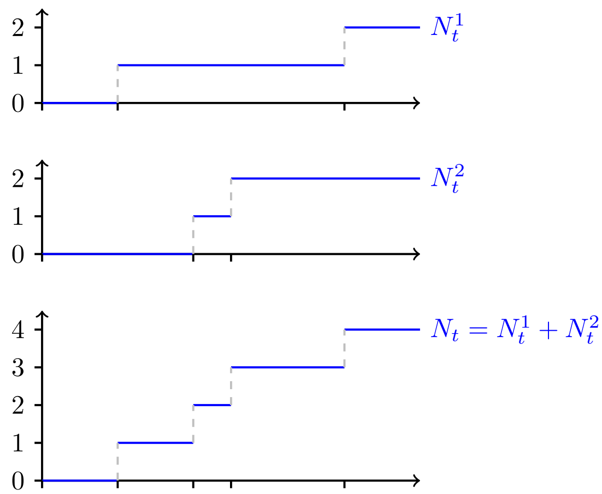
b. If $L < T$, the person dies from natural causes before he has the chance to get a heart attack. What is the probability that this happens?

Q. 2 (Smoking)**.** Two men, both 50 years old, lead independent lives. One is a nonsmoker, and his hazard rate function is h. The other is a smoker, and his hazard rate function is $2h$. What is the probability that the smoker outlasts the nonsmoker?

Q. 3 (Superposition and thinning)**.** Let $N_t^1, \ldots, N_t^k$ be independent Poisson processes with rates $\mu_1, \ldots, \mu_k$, respectively, and let $N_t = N_t^1 + N_t^2 + \cdots + N_t^k$.

a. What is $\mathbf{P}\{N_t = n\}$?

b. Each arrival of the process $\{N_t\}$ is an arrival of one of the processes $\{N_t^i\}$. Let us write $X_\ell = i$ if the ℓth arrival of $\{N_t\}$ is an arrival of $\{N_t^i\}$. For example, in the following figure we have $X_1 = 1$, $X_2 = 2$, $X_3 = 2$, $X_4 = 1$:



What is the distribution of the random variable X_ℓ?

c. What is the joint distribution of X_1 and X_2? Are they independent?

Q. 4 (Poisson gymnastics)**.** Let $\{N_t\}$ be a Poisson process with rate λ.

a. Compute $\mathbf{P}\{N_4 = 5 | N_9 = 13\}$.

b. For any $t, s \geq 0$, what is the conditional distribution of N_t given $N_s = n$?

c. Suppose each arrival of $\{N_t\}$ is a man or a woman with probability $\frac{1}{2}$. The genders of the arrivals are independent of each other and of the arrival times. Let M_t be the number of women that arrived by time t. Compute $\mathbf{E}(N_t | M_t = m)$.

Q. 5 (Buttons)**.** A monkey is presented with a red and a green button. The green button provides him with food, and the red button lets him use the bathroom. He repeatedly presses the buttons. Let X_k be the color of the kth button pressed by the monkey. Assume that $\{X_k\}$ are independent and $\mathbf{P}\{X_k = \text{red}\} = \frac{1}{3}$, $\mathbf{P}\{X_k = \text{green}\} = \frac{2}{3}$.

a. Let R_n be the number of times the monkey pressed the red button after pressing n buttons in total, and let G_n be the number of times the monkey pressed the green button. What is the joint distribution of R_n and G_n? Are R_n and G_n independent?

b. Let N be a Poisson random variable with mean a, independent of $\{X_k\}$. What is the joint distribution of R_N and G_N? Are R_N and G_N independent?